

Question 1: todo

Question 2: For the following proofs, let R be a commutative ring and $x, y, z \in R$.

- (a) *Proof.* We show that $(-x)(-y) + (-xy) = 0$, and thus $(-x)(-y) = xy$. As was shown in Proposition 5.7 in the notes, $-xy = (-x)y$. So, we compute that

$$(-x)(-y) + (-xy) = (-x)(-y) + (-x)y = (-x)(-y + y) = (-x) \cdot 0 = 0.$$

Thus, we are done. □

- (b)